Contextualized English Speaking Through AI-Supported Visual Concept Construction: A Case of Lunar New Year Market Narratives

Gloria Yi-Ming KAO*, Yun-Cheng CHOU
Graduate Institute of Digital Learning and Education, National Taiwan University of Science and Technology, Taipei 106, Taiwan, ROC
*gloriakao@gmail.com

Keywords: AI-supported language learning, oral proficiency, concept map, visual aids

Introduction

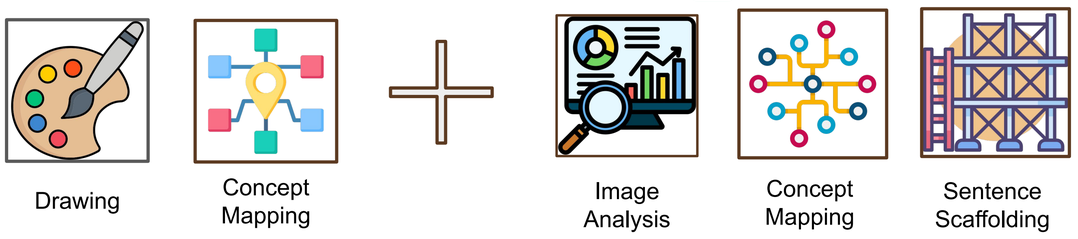
- **Many children hesitate to speak English** not only due to the fear of making grammer errors, more often, they simply have no idea what to describe in a given context.
- For example: 3 junior high school students were asked to describe their experience during their trip to Lunar New Year Market in Dadaocheng, all encountered the problem of not knowing what to say.



Objective

- This study aims to address this issue by incorporating graphic organizer with 3-stage AI-assisted scaffolding approach .
  - **Image Element Analysis**
  - **AI-Assisted Concept Mapping**
  - **Sentence Scaffolding**

Gemini
ChatGPT
assisted



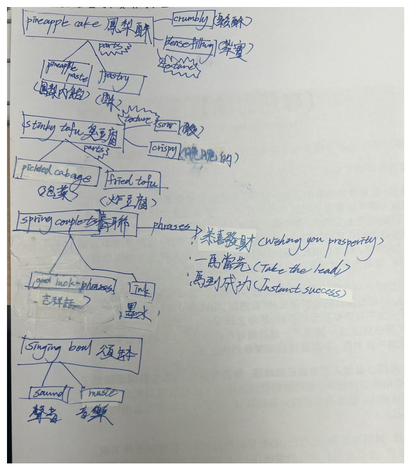
Pre AI-integration Task

- **STEP 1: Visualization via Hand Drawing**
  - The students were asked to draw the scenes that they intended to describe and to label the elements of their drawings with English keywords if possible.
  - Through visualization, the students could first express the narrative situations they intended to construct, including the identification of relevant characters, objects, and events.



Hand drawn visualization

- **STEP 2:Concept Mapping**
  - The students organized the elements in their drawings into concept maps
  - Concept maps were written in English, requiring students to indicate the relationships among concepts using English expressions.
  - This process aimed to support learners in organizing their ideas and establishing a coherent narrative framework prior to engaging in subsequent oral speaking activities.



Hand drawn concept map

3-Stage AI Assisted Scaffolding

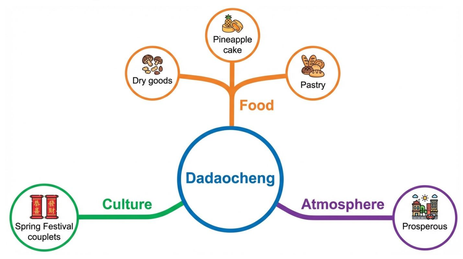
After students have finished drawing both images, AI was then introduced to support English-speaking practice in 3 stages.

- **STAGE 1:Image Element Analysis**
  - students uploaded their drawings for image analysis. The AI identified and annotated recognizable elements in the images with relevant English keywords
  - It also provided a list of these words along with translations. For instance, if a student had not written the English word for “旺,” ChatGPT might generate feedback such as “Prosperous (旺):meaning thriving or flourishing.
  - The students then practiced using these newly introduced or unfamiliar words.



AI generated anotations

- **STAGE 2:AI-Assisted Concept Mapping**
  - AI categorizes all identified vocabulary and generates a new concept map based on the elements recognized in the previous stage.
  - This allowed the students to see additional English words and phrases related to Dadaocheng beyond those in their original concept maps.



AI generated concept map

- **STAGE 3:Sentence Scaffolding**
  - AI provided example sentences relevant to the Dadaocheng context, such as: “Many visitors buy ‘pineapple cake,’ a popular Taiwanese souvenir, from the traditional bakeries in Dadaocheng.”
  - Students used these contextualized example sentences as models for English-speaking practice.



分類與情境例句 (Categorization & Contextual Sentences)

Goods (商品)

- Dry goods (乾貨)
  - *Example:* Dihua Street in Dadaocheng is famous for its wholesale **dry goods** stores, selling things like dried mushrooms and seafood.
- Spring Festival couplets (春聯)
  - *Example:* Before the Lunar New Year, the streets of Dadaocheng are filled with vendors selling red **Spring Festival couplets** for good luck.

Food (食物)

- Pastry (糕點)
  - *Example:* Some historic shops in Dadaocheng are well-known for their handmade traditional Taiwanese **pastry**.
- Pineapple cake (鳳梨酥)
  - *Example:* Many visitors buy **pineapple cake**, a popular Taiwanese souvenir, from the traditional bakeries in Dadaocheng.

Concepts / Adjectives (概念/形容詞)

- Prosperous (繁榮的)
  - *Example:* In the late 19th century, Dadaocheng was one of the most **prosperous** commercial areas in Taipei due to the tea trade.

AI generated example sentences

Result & Future Work

- The pilot study showed that students were able to describe the Lunar New Year atmosphere in Dadaocheng using richer vocabulary and more elaborated sentences. In the future, we will conduct a formal study to examine whether students using the proposed AI-assisted mechanism would demonstrate significantly better oral English-speaking ability and less oral speaking anxiety than those in the control group.
- With newer and more advanced commercial AI models, future studies will be able to utilize better multimodal capabilities of those models to achieve better image analysis and image generation results, thereby enabling more in-depth scaffolding.